

Packet Tracer - Design and Implement a VLSM Addressing Scheme

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Police	G0/0	10.1.1.97	255.255.255.240 /28	N/A
	G0/1	10.1.1.65	255.255.255.224 /27	N/A
	S0/0/0	10.1.1.121	255.255.255.252 /30	N/A
Schools	G0/0	10.1.1.113	255.255.255.248 /29	N/A
	G0/1	10.1.1.1	255.255.255.192 /26	N/A
	S0/0/0	10.1.1.122	255.255.255.252 /30	N/A
PD-1	VLAN 1	10.1.1.98	255.255.255.240 /28	10.1.1.97
PD-2	VLAN 1	10.1.1.66	255.255.255.224 /27	10.1.1.65
PS-101	VLAN 1	10.1.1.114	255.255.255.248 /29	10.1.1.113
PS-115	VLAN 1	10.1.1.2	255.255.255.192 /26	10.1.1.1
PD-1-11	NIC	10.1.1.110	255.255.255.240 /28	10.1.1.97
PD-2-23	NIC	10.1.1.94	255.255.255.224 /27	10.1.1.65
PS-101-87	NIC	10.1.1.118	255.255.255.248 /29	10.1.1.113
PS-115-12	NIC	10.1.1.62	255.255.255.192 /26	10.1.1.1

Objectives

In this lab you will design a VLSM addressing scheme given a network address and host requirements. You will configure addressing on routers, switches, and network hosts.

- Design a VLSM IP addressing scheme given requirements.
- Configure addressing on network devices and hosts.
- Verify IP connectivity.
- Troubleshoot connectivity issues as required.

Background / Scenario

You have been asked to design, implement, and test an addressing scheme for a customer. The customer has given you the network address that is suitable for the network, the topology, and the host requirements. You will implement and test your design.

Instructions

You have been given the network address **10.1.1.0/24** by your customer. The host address requirements are:

Note: Some of the information is provided in the Packet Tracer instructions for your topology.

Requirements

Host Requirements:

LAN	Number of Addresses Required
PD-1 LAN	11
PD-2 LAN	28

LAN	Number of Addresses Required
PS-101 LAN	5
PS-115 LAN	47
WAN	2

Design Requirements

- Create the addressing design. Follow guidelines provided in the curriculum regarding the order of the subnets.
- The subnets should be contiguous. There should be no unused address space between subnets.
- Provide the most efficient subnet possible for the point-to-point link between the routers.
- Document your design in a table such as the one below.

Subnet Description	Number of Hosts Needed	Network Address/CIDR	First Usable Host Address	Broadcast Address
PS-115 LAN [/26]	47	10.1.1.0	10.1.1.1	10.1.1.63
PD-2 LAN [/27]	28	10.1.1.64	10.1.1.65	10.1.1.95
PD-1 LAN [/28]	11	10.1.1.96	10.1.1.97	10.1.1.111
PS-101 LAN [/29]	5	10.1.1.112	10.1.1.113	10.1.1.119
WAN [/30]	2	10.1.1.120	10.1.1.121	10.1.1.123

Note: You will configure addressing on **all** devices and hosts in the network.

- Assign the first usable IP addresses in the appropriate subnets to Police for the two LAN links and the WAN link.
 - **G0/0 10.1.1.97,**
 - **G0/1 10.1.1.65,**
 - **S0/0/0 10.1.1.121**
- Assign the first usable IP addresses in the appropriate subnets to Schools for the two LANs links.
 - **G0/0 10.1.1.113,**
 - **G0/1 10.1.1.1**
- Assign the last usable IP address for the WAN link.
 - **S0/0/0 10.1.1.122**
- Assign the second usable IP addresses in the appropriate subnets to the switches.
 - **PD-1: 10.1.1.98,**
 - **PD-2 10.1.1.66,**
 - **PS-101 10.1.1.114,**
 - **PS-115 10.1.1.2**
- The switch management interface should be reachable from hosts on all of the LANs.
- Assign the last usable IP addresses in the appropriate subnets to the hosts.
 - **PD-1-11: 10.1.1.110,**
 - **PD-2-23 10.1.1.94,**
 - **PS-101-87 10.1.1.118,**
 - **PS-115-12 10.1.1.62**

If the addressing design and implementation are correct, all hosts and devices should be reachable over the network.